

Long-term Results of Iodine 125 Irradiation of Uveal Melanoma

Samuel Packer, MD,¹ Scott Stoller, MD,¹ Martin L. Lesser, PhD,²
Francine S. Mandel, PhD,² Paul T. Finger, MD¹

The authors report on 64 of the first 65 patients treated with iodine 125. The mean follow-up was 64.9 months. After treatment, 29 patients (45.3%) retained visual acuity of 20/100 or better, and 18 patients (28.1%) retained visual acuity within two lines of visual acuity before irradiation. Eleven patients (17.2%) died of metastasis, and 5 patients (7.8%) had local recurrence. Cataract developed in 29 (45.3%) patients; keratitis developed in only 2 (3.1%) patients, and dry eye developed in none. Neovascular glaucoma developed in 7 (10.9%) patients, and 15 (23.4%) patients had radiation retinopathy. Eleven patients (17.2%) required enucleation for either tumor growth or neovascular glaucoma. These results show the increasing number of radiation complications seen with long-term observation and the frequently seen adverse visual outcome.

Ophthalmology 1992;99:767-774

Iodine 125 is a low-energy radionuclide (35 keV) that has been used at North Shore University Hospital-Cornell University Medical Center since 1975.¹ We have used low-energy isotopes in gold scleral plaques (with sidewalls) to reduce radiation exposure to normal ocular structures and personnel. Our results are reported to update the follow-up of 64 of 65 patients on whom long-term data are available. Complications relate to death from metastasis and loss of vision due to radiation damage to normal ocular structures. This article will not compare these results with those from other radioactive sources, as this would be best done only with a prospective randomized trial or a statistical meta-analysis, which is beyond the scope of this report.

The rationale for using a low-energy isotope (¹²⁵I) versus a high-energy isotope (cobalt 60) is that normal ocular structures will receive less radiation with the former isotope. This relates to the inverse square law and the de-

crease in energy in tissue with distance from the radiation source. The rationale for using gold plaques as containers for the ¹²⁵I is that any metal with a high atomic number will maximally attenuate radiation. Therefore, gold scleral plaques with sidewalls will minimize radiation to the back and side of the plaque. Thus, we anticipated fewer ocular radiation complications with this technique. We expected no difference in survival, compared with other sources, because we were using an accepted radiation dose to the tumor (>80 Gy).

Patients and Methods

Sixty-four patients with choroidal melanoma were treated with ¹²⁵I plaque irradiation. All patients treated at our oncology center between February 1978 and January 1986 received this form of radiotherapy; after that date, we participated in the Collaborative Ocular Melanoma Study (COMS), and patients that were not entered into COMS were treated with other forms of therapy (e.g., thermoradiotherapy or palladium 103). The last follow-up visit reported was February 1991.

We have previously described our technique for ¹²⁵I gold plaque irradiation and have incorporated improvements in radiation dosimetry as they became available.² Since this technique (¹²⁵I/gold plaques) was first used in the United States in our center, errors in dosimetry that may have existed with the earlier patients must be accepted.

Originally received: October 15, 1991.

Revision accepted: December 13, 1991.

¹ Department of Ophthalmology, North Shore University Hospital-Cornell University Medical College, Manhasset.

² Department of Research/Division of Biostatistics, North Shore University Hospital-Cornell University Medical College, Manhasset.

Presented at the American Academy of Ophthalmology Annual Meeting, Anaheim, October 1991.

Reprint requests to Samuel Packer, MD, Department of Ophthalmology, North Shore University Hospital-Cornell University Medical College, 300 Community Dr, Manhasset, NY 11030.

The scleral plaques were 14 karat gold and had lips (raised edges). The plaque edge thickness varied from 0.5 to 1.0 mm. The attenuation of radiation from ^{125}I is greater than 95% with 0.1 mm of gold. The COMS design was adopted when it became available. The loading and treatment planning for radiation with multiple ^{125}I seeds (3-M, St. Paul, MN) was performed with computerized simulation (ADAC Lab, Milpitas, CA). After COMS protocols were available, they were followed for all patients.³ The only difference was that we chose 80 Gy as the dose to the apex of the tumor, as opposed to 100 Gy, the dose chosen for COMS patients. The dose rate varied from 40 to 120 cGy/hr. Except for several of the first six patients, where the plaques were only 1 mm larger than the largest basal tumor diameter, the plaque diameter was at least 2 mm greater than the largest basal dimension of the tumor.

The diagnosis of malignant melanoma of the choroid was made on all patients before treatment (by SP and/or PTF), using clinical examination and a variety of special tests. Visual acuity readings and complete eye examinations (including indirect ophthalmoscopy and slit-lamp biomicroscopy) were performed on both eyes. Special tests included ultrasonography (A and B scans), fluorescein angiography, and complete systemic evaluation. Routine tests included chest roentgenography and liver function studies. No scans were performed unless indicated by initial evaluation. After the COMS protocol was available, it was followed for all patients.

Tumor size was established initially with indirect ophthalmoscopy and ultrasonography, and, when it became available, we followed the COMS protocol. Tumors were considered small if their height was less than or equal to 3 mm and/or the base diameter was less than or equal to 8 mm, medium if the height was 3.1 to 8.0 mm and/or the base diameter was 8.1 to 16.0 mm, and large if the height was greater than or equal to 8.1 mm and/or the base diameter was greater than or equal to 16.1 mm. Tumor location was considered anterior if more than 50% of the diameter of the tumor was anterior to the equator and posterior if more than 50% of the diameter the tumor was posterior to the equator. No patient was treated who had a tumor height less than 2.0 mm or less than 6.0 mm in base diameter.

Patients were seen in follow-up at North Shore University Hospital-Cornell University Medical Center (by either SP or PTF). Initially, patients were seen at 6-month intervals (for 2 years), then yearly. Patients with complications were seen more often. Those patients who were seen by referring physicians had copies of relevant tests sent to North Shore University Hospital-Cornell University Medical Center. When necessary, we used this material or telephoned for needed information. No patients were lost to follow-up, however there were inadequate survival data for one patient and for visual data for three patients. Of the 64 patients studied, 60 were followed for 2 years or longer, 50 were followed for longer than 3 years, 41 were followed for longer than 4 years, and 34 were followed for longer than 5 years.

Visual acuity was measured in all patients at each follow-up visit (not necessarily best corrected). Most patients

had refractions with their initial visit and then at approximately yearly intervals. Yearly tests included complete eye examination, fluorescein angiography, ultrasonography and complete blood count, liver profile, and serum alanine aminotransferase.

The following variables were studied: age, sex, eye involved, presence or absence of diabetes mellitus, hypertension, cataract, retinal detachment, vitreous hemorrhage, symptoms, tumor diameter, tumor height, tumor location, distance from optic nerve, distance from fovea, chemotherapy after radiation, photocoagulation, rubeosis, neovascular glaucoma, radiation retinopathy, macular edema, tumor metastasis, local recurrence, and performance of enucleation.

Eleven patients received chemotherapy. This consisted of carmustine (BCNU) and vinblastine (8 patients, 1 patient experienced nausea after receiving BCNU and was switched to lomastine [CCNU]). Two patients received BCNU and dacarbazine (DTIC), and one patient received only DTIC.

Statistical Methods

End Point Variables

Three end point variables were examined: survival from date of diagnosis until death, time from diagnosis until metastasis, and time from diagnosis until the occurrence of an adverse visual event. An adverse visual event was defined as either visual acuity of 20/100 or worse or losing 2 or more lines of visual acuity. Losing 2 or more lines of visual acuity was defined as having visual acuity decrease two (or more) lines on a standard eye chart (e.g., 20/20 to 20/40).

Statistical Analysis

Univariate survival analyses were performed to determine which of the independent variables noted above were predictors of the three end point variables. The survival distributions were estimated using the Kaplan-Meier product limit method. Hypotheses for differences between the groups were tested using the log-rank test. After the significant ($P \leq 0.05$) variables were identified, Cox regression analysis was used to identify the significant predictors of time until the end point events. A backward selection technique was used to eliminate any variables that did not significantly add to the prediction.

Results

We analyzed our data to assess the probability of an adverse visual event. An adverse visual event was either visual acuity decreasing to 20/100 or worse or visual acuity decreased 2 lines or more. Table 1 lists the variables: age, sex, diabetes mellitus (7 patients), hypertension (15 patients), presence of cataract before treatment (10 patients), right eye (36 patients) versus left eye, presence of symptoms before treatment (49 patients), tumor diameter (31

Table 1. Univariate Analysis, Adverse Visual Event

Variable	No.	P Value Log-Rank Chi-square	Variable	No.	P Value Log-Rank Chi-square
Diabetes Mellitus		0.0079	Macular Edema		0.3413
Yes	7		No	46	
No	55		Yes	16	
Tumor Diameter		0.0469	Radiation Retinopathy		0.3457
Small	31		No	47	
Medium/large	31		Yes	15	
Rubeosis Irides		0.0895	Distance from Fovea		0.4174
No	50		0-1.0 mm	15	
Yes	12		1.1-3.0 mm	19	
Metastasis		0.0982	>3.0 mm	28	
No	52		Preoperative Vitreous Hemorrhage		0.4289
Yes	10		No	56	
Preoperative Retinal Detachment		0.1170	Yes	6	
No	25		Symptoms		0.4356
Yes	37		Yes	48	
Distance from Optic Nerve		0.1266	No	14	
0-1.0 mm	10		Age		0.5144
1.1-3.0 mm	18		0-59 yrs	24	
>3.0 mm	34		Older than 60 yrs	38	
Hypertension		0.1485	Postoperative Cataract		0.5965
Yes	15		No	26	
No	47		Yes	36	
Sex		0.1615	Eye		0.6335
Male	26		Right	35	
Female	36		Left	27	
Postoperative Chemotherapy		0.2274	Enucleation		0.6436
No	51		No	51	
Yes	11		Yes	11	
Postoperative Photocoagulation		0.2470	Tumor Location		0.6674
No	57		Anterior to equator	14	
Yes	5		Posterior to equator	47	
Tumor Height		0.2874	Neovascular Glaucoma		0.7862
Small/medium	34		No	55	
Large	28		Yes	7	
Postoperative Vitreous Hemorrhage		0.3114	Preoperative Cataract		0.9990
No	48		Yes	10	
Yes	14		No	52	

small, 28 medium, and 3 large), tumor height (3 small, 31 medium, and 28 large), location of tumor (anterior, 14 patients) versus posterior to equator, distance from optic nerve (10 patients, 0 to 1 mm; 18 patients, 1.1 to 3.0 mm; and 34 patients, >3.0 mm), distance from fovea (15 patients, 0 to 1.0 mm; 19 patients, 1.1 to 3.0 mm, and 28 patients, >3.0 mm), presence of retinal detachment

before treatment (37 patients), presence of vitreous hemorrhage before treatment (6 patients), whether the patient received chemotherapy in the postoperative course (11 patients), photocoagulation after irradiation (5 patients), cataract after irradiation (36 patients), rubeosis iridis (12 patients), neovascular glaucoma (7 patients), vitreous hemorrhage after irradiation (14 patients), radiation ret-

inopathy (15 patients), macular edema (16 patients), melanoma metastasis (10 patients), and local recurrence (5 patients). Two observations were missing values for exact tumor height, diameter, and exact location. Therefore, these patients were eliminated from pertinent analyses.

Thus, in Table 1, the number of patients reported is 61 or 62; whereas in Table 2 we report on 64 observations except for 1 patient without adequate information for tumor location and 2 patients for distance to fovea.

Thirty-one (50%) patients had a small (0 to 8 mm)

Table 2. Univariate Analysis Time to Metastasis

Variable	No.	P Value Log-Rank Chi-square	Variable	No.	P Value Log-Rank Chi-square
Cataract after Irradiation		0.0369	Age		0.4871
Yes	37		>60 yrs	40	
No	27		<60 yrs	24	
Chemotherapy		0.0586	Enucleation		0.5451
Yes	12		Yes	11	
No	52		No	53	
Eye		0.0633	Cataract before Irradiation		0.5783
Right	36		Yes	10	
Left	28		No	54	
Location		0.0862	Symptoms		0.6564
Anterior	14		Yes	49	
Posterior	49		No	15	
Macular Edema		0.1696	Vitreous Hemorrhage after Irradiation		0.6577
No	47		Yes	15	
Yes	17		No	49	
Tumor Height		0.2461	Hypertension		0.6871
Large	29		Yes	15	
Small/medium	35		No	49	
Distance to Optic Nerve		0.2475	Retinal Detachment		0.7349
0-1 mm	11		Yes	38	
1.1-3.0 mm	18		No	26	
>3.0 mm	35		Sex		0.7646
Vitreous Hemorrhage		0.2722	Male	28	
Yes	7		Female	36	
No	57		Distance to Fovea		0.8373
Photocoagulation		0.3369	0-1.0 mm	15	
Yes	5		1.1-3.0 mm	20	
No	59		>3.0 mm	27	
Local Recurrence		0.4723	Neovascular Glaucoma		0.8952
No	56		Yes	7	
Yes	8		No	57	
Tumor Diameter		0.4758	Rubeosis		0.9052
Medium/large	33		Yes	12	
Small	31		No	52	
Radiation Retinopathy		0.4811			
No	48				
Yes	16				
Diabetes Mellitus		0.4871			
Yes	7				
No	57				

tumor diameter. Thirty-one patients had medium or large tumor diameters (>10 mm). Those with small tumors had a significantly smaller ($P = 0.0469$) rate of adverse visual event (Table 1).

Time Until Death

Thirty-six patients (56%) had cancer in the right eye. Patients with right eye tumors had poorer survival ($P = 0.0633$) than those with disease in the left eye (Table 2).

Twelve patients (19%) received chemotherapy. Patients who received chemotherapy had significantly poorer survival ($P = 0.0058$) than those who did not receive chemotherapy (Table 1).

The Cox regression analysis to predict time until death was calculated using left or right eye and chemotherapy as predictors. Metastasis was not included in this analysis for two reasons. First, all patients who died of disease had a metastasis, and, therefore, including metastasis would obscure the relationship between survival and the other predictors. Second, because the method was to use variables known at time of diagnosis and metastasis was not a presenting variable, it was not considered to be clinically appropriate for this model. The backward selection technique eliminated left or right eye from the model, leaving chemotherapy as the sole predictor of time until death. The risk of death, as expressed by the odds ratio, was 3.69 times greater for those who received chemotherapy (1.21 to 14.52, 95% confidence interval) (Table 3).

Time Until Metastasis

The time until a metastasis was discovered was analyzed, because in this study having a metastasis was almost synonymous with dying of disease. One patient in whom a metastasis developed was excluded from this analysis because the date of metastasis was not available. No variables were found to predict a metastasis.

Time Until Adverse Visual Event

Forty-nine (77.8%) of 63 patients had an adverse visual event (1 patient of the 63 was censored because inadequate information was available). History of diabetes was associated with experiencing an adverse event. Seven patients (11.3%) had diabetes. Patients having a history of diabetes had a significantly higher ($P = 0.0079$) rate of an adverse visual event than those without diabetes (Table 1).

At 2 years, 33 (55%) of 60 (4 died) had an adverse visual event; at 3 years, 31 (62%) of 50 had an adverse

Table 4. Cox Regression for Adverse Visual Event

	<i>P</i> Value (Chi-square)	Risk Ratio
Diabetes	0.0091	2.942
Tumor diameter	0.2024	0.922

visual event; at 4 years, 28 (68.3%) of 41 had an adverse visual event; and, at 5 years, 25 (73.5%) of 34 had an adverse visual event.

Cox regression analysis to predict time until an adverse visual event using diabetes ($P = 0.0079$) and largest tumor diameter ($P = 0.0469$) was computed. Diabetes was the only significant predictor of time until an adverse visual event in the Cox regression model. The risk of an adverse visual event was 2.942 greater for those who had diabetes (1.27 to 6.37, 95% confidence interval) (Table 4). A univariate analysis of adverse visual event related the following post-treatment events: retinal detachment ($P = 0.04$), rubeosis irides ($P = 0.07$), and vitreous hemorrhage ($P = 0.10$).

Discussion

Univariate analysis (Table 2) of time to metastasis showed a P value of 0.0369 for patients with cataract after treatment. However, this variable lost its significance with Cox regression analysis (Table 3).

These results indicate that death of metastatic melanoma after irradiation with ^{125}I was significantly related to the use of chemotherapy after irradiation. Only patients who had multiple poor prognostic indications (large base diameters [>16.1 mm], large tumor height [>8.1 mm], and/or anterior location) were selected for chemotherapy before clinical evidence of metastasis. Thus, it is no surprise that this variable was significantly related. More surprising is the lack of any significant association of death of disease with variables that other investigators have found to be significant (tumor diameter, tumor height, anterior location, local tumor recurrence).⁴⁻⁷ Among the explanations are that there were only 11 patients who died of disease (too small a cohort) and that our follow-up was too short. Finally, multivariate analysis (Table 3) showed that patients had a worse prognosis if the melanoma was in the right eye. No explanation is offered, although metastatic pattern may vary with location, as has been shown by previous research.^{8,9}

The fact that adverse visual outcome was significantly related to diabetes mellitus (P value, 0.0079) and tumor diameter (P value, 0.0469) is not surprising. The cause of decreased vision was frequently due to radiation retinopathy and/or maculopathy, both of which may be due in part to the increased susceptibility of the diabetic vasculature to radiation damage. The larger tumor diameter may be related to poor visual outcome since a larger area of the normal retina was irradiated and thus a higher probability existed of irradiating the macula with damaging doses.

Table 3. Cox Regression for Death of Metastasis

	<i>P</i> Value (Chi-square)	Risk Ratio
Eye	0.1234	0.189
Chemotherapy	0.0733	3.169

The incidence of cataract seemed high, and, therefore, we analyzed this further. Ten patients (15.6%) had cataract before irradiation. Thirty-seven (56.3%) patients who had cataract after irradiation were analyzed to determine whether the cause of visual loss was due to a radiation-induced cataract or to an aging lens (nuclear sclerosis). Of the 15.6% of patients with cataracts before irradiation, only 1 patient had a posterior subcapsular cataract that was progressive and contributed to the loss of vision. In addition, 6 patients were noted to have nuclear sclerotic cataract after treatment, and 8 patients had cortical spoke cataracts noted after treatment—these cataracts may represent a normal aging of the lens or accelerated aging caused by radiation. Thus, although cataracts were noted after radiation in 56.3% of patients, only 45.3% were believed to be related to radiation. This analysis is further complicated by the fact that in those cases with visual loss, multiple radiation complications occurred that may have been the cause of visual decline (in addition to the presence of cataract), and it was difficult to assign cause. Thus, in only 6.3% of cases were we able to state clearly that the loss of vision was due to a radiation-induced cataract.¹⁰

Our original intent with introducing a low-energy isotope (¹²⁵I) was to determine if fewer complications would result than with the use of a high-energy isotope (cobalt 60).¹¹ The basis for this was simply that the inverse square law indicates that a high-energy and a low-energy isotope each deliver the same amount of radiation to the apex of a choroidal melanoma. Therefore, the normal ocular structures would always receive less radiation with a low-energy source. Several estimates have been made of radiation dose distribution, but roughly 10% to 30% less radiation is delivered to normal ocular structures (with a low-energy isotope), depending on their location relative to the tumor.^{12,13} From this observation, one would assume that a clinical trial would merely confirm this. No prospective randomized clinical trial has been performed to compare low-energy to high-energy radiation therapy of choroidal melanoma, nor has such a study been performed with any two forms of radiation (including charged particles). Therefore, short of the statistical technique of a meta-analysis of reports, there is no honest way of comparing one technique with another.

Also, a frequently overlooked point is that all the existing techniques—whether plaque or charged particle—are local treatments. That is, they are likely to have the same local control while having no effect on metastasis (if it is present at the time of treatment). Thus local control is possible with any source of radiation (if the same dose and dose rate is used), and, therefore, the issue should be (for now) the frequency of ocular complications. If complication rates are similar, than issues of cost and convenience may be relevant. Thus, readers will have to draw their own conclusions with regard to the comparative effectiveness, using the data presented here and the data presented by others.

Our follow-up data (relative long-term) indicate that the eye encounters significant morbidity (loss of vision) when irradiated with ¹²⁵I. This is primarily due to the

dose of radiation received by normal ocular structures and relates to the tumor apex dose that is chosen (i.e., the higher the tumor apex dose, the higher the dose to normal ocular structures and, therefore, the greater probability of radiation damage). Local control was good (92.2%), and retention of eye was significant (82.8%). It should be noted that we had a high tumor apex dose (mean, 91 Gy), and thus complications may have been greater than if we had chosen a lower dose of 55 Gy (see below).

Our rate of regrowth may have been adversely influenced by the original plaques (first 6 cases) being designed to be only 1 mm greater in diameter than the base diameter of the tumor. This, combined with the fact that plaque placement may be more difficult with posterior tumors, led to a greater-than-expected frequency of regrowth. This explanation is partially supported by the fact that other groups using ¹²⁵I have not had a high frequency of tumor regrowth.^{14,15}

We strongly believe that improvements are needed to further reduce radiation to normal ocular structures. Several authors have elected to lower the dose to the tumor, which, of course, lowers the dose to normal ocular structures and will reduce radiation complications.^{4,16} The problem is that the tumoricidal radiation dose for choroidal melanoma is not known.^{17,18} Thus, if one were to equate melanoma with sarcoma or use skin melanoma radiation sensitivity, one would not deliver less than 55 Gy to the tumor apex. The use of a relative biologic effectiveness factor with charged particles (1.1 for protons, 1.3 for helium ions) is partially philosophically based. One could either use 55 Gy and have the added benefit of a relative biologic effectiveness factor or decrease the dose by using a relative biologic effectiveness factor.¹⁹ The latter clearly would reduce complications but has the risk of increased mortality. Iodine 125 also has been shown to have a relative biologic effectiveness factor greater than 1.0.²⁰ However, we have chosen not to use this clinically to lower the radiation dose; thus, our mean tumor apex dose of greater than 90 Gy with ¹²⁵I versus the less than 80 GyE with charged particles may not be equivalent clinically.⁴⁻⁶ It is important to note that a total dose to tumor of less than 50 Gy has been associated with significant treatment failure.¹⁷

Because of the significant complication rate noted with ¹²⁵I, our oncology service is currently using palladium 103 (27 keV) to further lower the radiation to normal ocular structures. Also, for the past 6 years, we have treated patients not eligible for the Collaborative Ocular Melanoma Study³ with low-energy isotopes (¹²⁵I or palladium 103) and microwave hyperthermia.²¹ When using this technique (thermoradiotherapy), we have lowered the radiation dose to 55 Gy. The rationale here is that radiation and heat work synergistically to kill cancer cells.²² This approach to lowering the radiation dose to the tumor has strong clinical support, as opposed to lowering the dose because of the possible clinical benefits of a relative biologic effectiveness factor with charged particles.

Long-term results are necessary to properly appraise any new forms of therapy. Low-energy brachytherapy

(^{125}I)^{14,15,23} and charged particle therapy^{4,5} have been in use for approximately 15 years.^{24,25} The visual results with ^{125}I (27.7% retained visual acuity within 2 lines of preoperative levels, and visual acuity decreased to less than 20/100 in 44.6%) compares favorably with the visual results with charged particles. Lindstadt et al⁵ report their visual results after helium ion therapy as 38% retaining visual acuity to within 2 lines of preoperative levels and 53% with visual acuity decreasing to 20/200 or worse; similarly Gragoudas et al^{4,6} report that, after proton beam therapy, visual acuity will decrease to 20/200 or worse in 42% of patients. Thus, visual outcome appears similar, and death from disease also shows similarities: ^{125}I , 17.2%; helium ions, 13.4%,⁵ and proton beam, 14%.^{4,6} Comparison of both visual results and mortality is important, and we hope that this article facilitates that comparison. We recognize that these comparisons are faulty since they are not from comparable cases with regard to tumor size, location, or length of follow-up. Nor are they comparable with regard to radiation dose. For instance, our mean radiation dose to tumor apex was 90 Gy, whereas Lindstadt et al used tumor apex doses between 50 and 80 GyE. They used 50 Gy for all posterior tumors. This is surprising, since they had noted regrowth with 70 GyE and therefore chose 80 GyE for all other tumors (except for a clinical trial at various doses). A report by Karlsson et al²⁶ revealed that patients treated with cobalt 60 who had tumor growth after irradiation had a greater chance of dying of metastases than did patients whose melanomas did not grow. Thus, reducing dose may have significant consequences! This concern may be even more noteworthy if relative biologic effectiveness factor estimates are not clinically valid. Therefore, lowering tumor radiation dose to reduce radiation damage to normal ocular structures must be carefully considered because of the potential of increasing mortality.

References

- Packer S, Rotman M. Radiotherapy of choroidal melanoma with iodine-125. *Ophthalmology* 1980;87:582-90.
- Packer S, Fairchild RG, Salanitro P. New techniques for iodine-125 radiotherapy of intraocular tumors. *Ann Ophthalmol* 1987;19:26-30.
- Earle J, Kline RW, Robertson DM. Selection of iodine 125 for the Collaborative Ocular Melanoma Study. *Arch Ophthalmol* 1987;105:763-4.
- Gragoudas ES, Egan KM, Seddon JM, et al. Survival of patients with metastases from uveal melanoma. *Ophthalmology* 1991;98:383-90.
- Lindstadt D, Castro J, Char D, et al. Long-term results of helium ion irradiation of uveal melanoma. *Int J Radiat Oncol Biol Phys* 1990;19:613-18.
- Gragoudas ES, Seddon JM, Egan K, et al. Long-term results of proton beam irradiated uveal melanomas. *Ophthalmology* 1987;94:349-53.
- Augsburger JJ, Gamel JW. Clinical prognostic factors in patients with posterior uveal malignant melanoma. *Cancer* 1990;66:1596-1600.
- Albert DM, Rubenstein RA, Scheie HG. Tumor metastasis to the eye. I. Incidence in 213 adult patients with generalized malignancy. *Am J Ophthalmol* 1967;63:723-6.
- Fidler IJ, Poste G. The heterogeneity of metastatic properties in malignant tumor cells and regulation of the metastatic phenotype. In: Owens AH Jr, Coffey DS, Baylin SB, eds. *Tumor Cell Heterogeneity: Origins and Implications*. New York: Academic Press, 1982;127-45.
- Merriam GR Jr, Focht EF. A clinical study of radiation cataracts and the relationship to dose. *AJR Am J Roentgenol* 1957;77:759-85.
- Packer S, Rotman M, Fairchild RG, et al. Irradiation of choroidal melanoma with iodine-125 ophthalmic plaque. *Arch Ophthalmol* 1980;98:1453-7.
- Chiu-Tsao S-T, O'Brien K, Sanna R, et al. Monte Carlo dosimetry for ^{125}I and ^{60}Co in eye plaque therapy. *Med Phys* 1986;13:678-82.
- Astrahan MA, Luxton G, Jozsef G, et al. An interactive treatment planning system for ophthalmic plaque radiotherapy. *Int J Radiat Oncol Biol Phys* 1990;18:679-87.
- Garretson BR, Robertson DM, Earle JD. Choroidal melanoma treatment with iodine 125 brachytherapy. *Arch Ophthalmol* 1987;105:1394-7.
- Lean EK, Cohen DM, Liggett PE, et al. Episcleral radioactive plaque therapy: initial clinical experience with 56 patients. *Am J Clin Oncol* 1990;13:185-90.
- Kindy-Degnan NA, Char DH, Castro JR, et al. Effect of various doses of radiation for uveal melanoma on regression, visual acuity, complications, and survival. *Am J Ophthalmol* 1989;107:114-20.
- Overgaard J, Overgaard M, Hansen PV, van der Maase H. Some factors of importance in the radiation treatment of malignant melanoma. *Radiation Oncol* 1986;5:183-92.
- Trott KR. The optimal radiation dose per fraction for the treatment of malignant melanomas [editorial]. *Int J Radiat Oncol Biol Phys* 1991;20:905-7.
- Castro JR. Heavy charged particle irradiation of human cancers. *Radiat Med* 1983;1:70-5.
- Marchese MJ, Goldhagen PE, Zaider M, et al. The relative biological effectiveness of photon radiation from encapsulated iodine-125, assessed in cells of human origin: I. normal diploid fibroblasts. *Int J Radiat Oncol Biol Phys* 1990;18:1407-13.
- Finger PT, Packer S, Paglione RW, et al. Thermoradiotherapy of choroidal melanoma. Clinical experience. *Ophthalmology* 1989;96:1384-8.
- Hahn GM. *Hyperthermia and Cancer*. New York: Plenum Press, 1982;56-85.
- Billson FA, Painter G, Dunlop I, et al. The conservative management of choroidal melanoma. *Aust N Z J Ophthalmol* 1990;18:65-75.
- Lommatzsch PK. Results after β -irradiation ($^{106}\text{Ru}/^{106}\text{Rh}$) of choroidal melanomas: 20 years' experience. *Br J Ophthalmol* 1986;70:844-51.
- Foerster MH, Bornfeld N, Schulz U, et al. Complications of local beta radiation of uveal melanomas. *Graefes Arch Clin Exp Ophthalmol* 1986;224:336-40.
- Karlsson UL, Augsburger JJ, Shields JA, et al. Recurrence of posterior uveal melanoma after ^{60}Co episcleral plaque therapy. *Ophthalmology* 1989;96:382-8.

Discussion
by
Dennis M. Robertson, MD

Following the pioneering work of Professor Sealy, who, with his associates, reported treatment of a malignant uveal melanoma with iodine 125 15 years ago, Dr. Samuel Packer became the first American ophthalmologist to use ^{125}I in the management of uveal melanoma. Four years ago, I discussed Dr. Packer's preliminary observations of 28 eyes treated with ^{125}I brachytherapy at North Shore University Hospital in New York. As would be expected with short follow-up of a small number of patients, the complications reported were few. However, Dr. Packer was correct when he predicted that radiation complications would, with longer follow-up, "be substantial." Let us review the current study.

Of 64 patients treated with ^{125}I having a mean follow-up of approximately 5 years, almost half were legally blind, approximately half had cataracts, approximately a fourth had radiation retinopathy, approximately a fifth required enucleation because of tumor growth or neovascular glaucoma, and approximately a fifth of the patients were dead. It is to be noted that, although the mean follow-up was 5 years, more than a fifth of the patients were followed less than 3 years, so these numbers do not accurately represent the actual complications one would see at a 5-year follow-up; therefore, the reported complications here are probably under-represented. Unfortunately, it appears that the radiation complications and failures with ^{125}I do not differ significantly from the radiation complications associated with other sources of radiation, such as charged particles, radioactive gold, or even cobalt 60.

There are two key issues in the management of uveal melanomas. The first is patient survival. The second is preservation of vision. The Collaborative Ocular Melanoma Study is looking at factors that influence survival. One of the important questions being asked by the Collaborative Ocular Melanoma Study is

whether patient survival is better with enucleation or radiation (specifically radiation with ^{125}I). Recruitment in that study at this time is very encouraging, and, with continued support from ophthalmologists in North America, I think an answer to that question will be forthcoming.

The second key issue in the management of uveal melanoma, preservation of vision, continues to assume great importance and will be even more important if the Collaborative Ocular Melanoma Study determines that radiation is equal to or better than enucleation in terms of patient survival. Considering that almost half of the eyes treated with radiation become legally blind within 5 years, we must acknowledge that we are doing a less than optimal job of preserving vision when we manage ocular melanomas with radiation. It seems appropriate, then, that research efforts should be directed toward learning ways to reduce the complications of radiation. While there continue to be some studies investigating different sources of radiation and others investigating certain drugs or hyperthermia that can be used synergistically with radiation, we still know too little to recommend these approaches for general clinical use. We must remember that for any modification of therapy for melanoma, there probably is a period of 5 or more years before the true value of the treatment can be determined. Anecdotal claims, therefore, can always be made a few steps ahead of good science. At this time, I think it is especially important that quality research be carried out in the laboratory to learn ways that we can reduce the complications of radiation. This area of research deserves high priority.

References

1. Sealy R, le Roux PLM, Rapley F, et al. The treatment of ophthalmic tumours with low-energy sources. *Br J Radiol* 1976;49:551-4.
2. Packer S. I-125 radiation of posterior uveal melanoma. *Ophthalmology* 1987;94:1621-6.

From the Department of Ophthalmology, Mayo Clinic, Rochester.